DS3000 Stock ESG and Valuation Analysis

Presentation by Hang Hang, Sahaj Baxi, Joshua Sharma, Beatriz Braz Auto Marinho

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Introduction

- Investigated the relationship between ESG ratings and company performance.
- Analyzed real-time stock data, ESG scores, and financial metrics using machine learning.
- Some key insights:
- ESG alone doesn’t predict performance but reveals patterns when combined with financial data.
- States the complexity of stock valuation which depends on multiple factors.

Data Description

ESG Data

- ESG Score
- Individual environmental, social, and governance scores
- ESG descriptor - categorical (negligible, medium, or severe risk)
- Data taken from Yahoo Finance ESG Library

Stock Data

- Real-time stock data analyzing stock price over a period of time
- Market capitalization measures overall company valuation based on outstanding shares times share price

Financial Data

- Beta measures a stock's volatility compared to the market
- PE ratio (price-to-earnings ratio)
- EPS TTM (Earnings Per Share - Trailing Twelve Months)
- Data taken from Yahoo Finance Libraries

Data Sources

- Used Yahoo Finance's API to retrieve live ESG scores
- Used Python libraries with Slickcharts.com to retrieve the financial metrics of the top 100 S&P 500 companies

Data Relationship And Purpose

We leveraged Stock Data and Financial Data to quantify a companies performance. We used ESG Data to describe the companies ESG score through the numeric score and subscores, and through the categorical descriptor of risk level. Given this data we were able to leverage machine learning to make conclusions on the relationship between ESG and company performance.

Polynomial Regression Methods

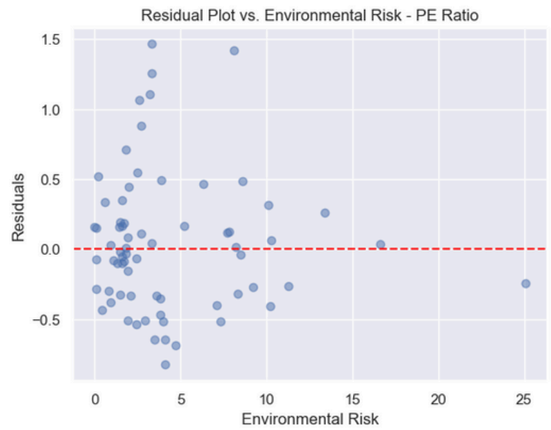
- Introduced quadratic and interaction terms (Environmental × Social) to capture non-linear relationships.
- Applied log-transform to Price for better modeling.
- Evaluated models using 70/30 train-test split and Leave-One-Out Cross-Validation (LOO-CV).

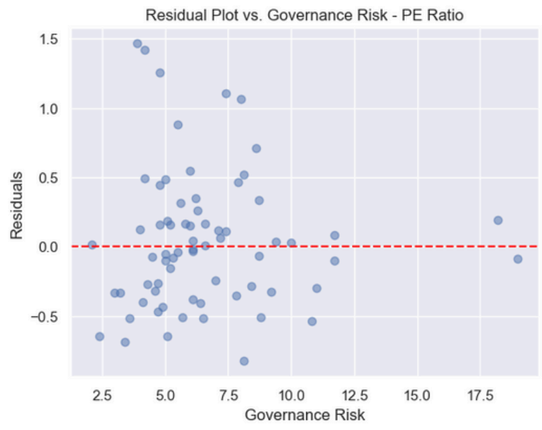
Target	Model Type	MSE (LOO-CV)	R^2 (LOO-CV)
Price	Linear	High (427292)	Negative (-0.0251)
Price	Multiple (Using ESG Subscores as Predictors)	Low (0.9398, likely due to log transform)	Negative (-0.06)
Price, Market Cap, PE Ratio	Polynomial (Use ESG Subscores + Env x Soc Interaction)	Lowest (For PE Ratio: 0.2660; Singlefold CV for Market Cap: 0.3238)	Higher (For PE Ratio: 0.1364; Singlefold CV for Market Cap: 0.1577)

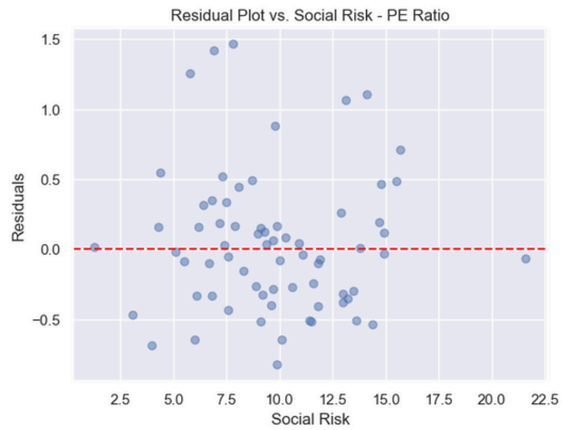
```python
# Add interaction term: Environmental Risk * Social Risk
interaction_term = (consolidated_df['Environmental Risk'] * consolidated_df['Social Risk']).to_numpy()
X_interaction = np.hstack([X, interaction_term.reshape(-1, 1)])

# Update feature names to include the interaction term
updated_features = features + ['Env_Soc_Interaction']
print("Features with Interaction Term:", updated_features)
```

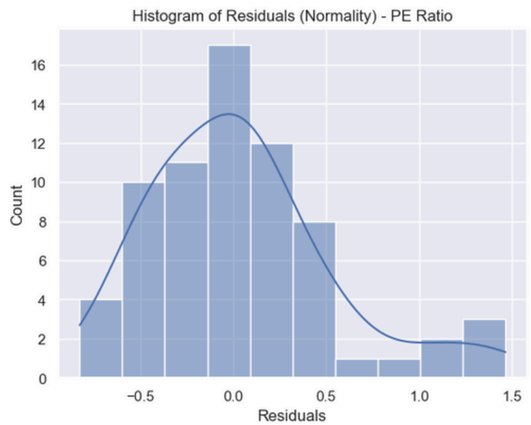
Polynomial Regression Results

```
Target: Price
Single-fold CV MSE: 0.7801199862002741
Single-fold CV R^2: -0.2144072832106263

Target: Market Cap (Billion)
Single-fold CV MSE: 0.3238354655403183
Single-fold CV R^2: 0.1577453902340239

Target: PE Ratio
Single-fold CV MSE: 0.33983552349208296
Single-fold CV R^2: 0.030141222086304653

Target: Price
LOO-CV MSE: 0.9596139319757165
LOO-CV R^2: -0.08265336071113993

Target: Market Cap (Billion)
LOO-CV MSE: 0.7059867517885312
LOO-CV R^2: -0.0899000161919612

Target: PE Ratio
LOO-CV MSE: 0.26601493619896494
LOO-CV R^2: 0.1364099221004319
```









- Price Model: High MSE, negative R^2, minimal improvement over linear models.
- Market Cap Model: Improved single-fold R^2 with polynomial terms.
- PE Ratio Model: Promising R^2 from LOO-CV, indicating some non-linear relationship might be captured.
- Residual analysis shows some normality for the social risk and PE Ratio model but shows a slight funnel shape for the env risk and gov risk, indicating heteroscedasticity

PCA-Based Clustering

Methods

- Dimensionality Reduction: Applied PCA to condense ESG-financial dimensions into a smaller set of principal components, retaining over 90% of original variance
- Clustering Approach: K-Means clustering (k=4) applied to PCA-transformed data, enabling identification of company groups with distinct ESG and financial profiles

```
numerical_features = [
    'Total ESG Risk Score', 'Environmental Risk', 'Social Risk', 'Governance Risk',
    'Market Cap (Billion)', 'Beta', 'PE Ratio', 'EPS TTM', 'Price', 'HPR']
categorical_features = ['ESG Description']
```

```
# Filter numeric features
X_numeric = consolidated_df[numerical_features].select_dtypes(include=['float64', 'int64']).dropna()

# Encode categorical features
encoder = OneHotEncoder(sparse_output=False, drop='first')
X_categorical = encoder.fit_transform(consolidated_df[categorical_features].astype(str).fillna('Unknown'))

# Combine numeric and encoded categorical features
X_categorical_df = pd.DataFrame(
    X_categorical,
    columns=encoder.get_feature_names_out(categorical_features),
    index=consolidated_df.index)

# Combine numeric and encoded categorical features
X_combined = pd.concat([X_numeric, X_categorical_df], axis=1)

# Standardize combined features to ensures they have a mean of 0 and standard deviation of 1.
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_combined)

# Apply PCA for dimensionality reduction
pca = PCA(n_components=3)
X_pca = pca.fit_transform(X_scaled)

# Create DataFrame for PCA components
pca_df = pd.DataFrame(X_pca, columns=['PCA1', 'PCA2', 'PCA3'])
pca_df['Ticker'] = consolidated_df.index

# Perform KMeans clustering
optimal_k = 4
kmeans = KMeans(n_clusters=optimal_k, random_state=42)
clusters = kmeans.fit_predict(pca_df[['PCA1', 'PCA2', 'PCA3']])
```

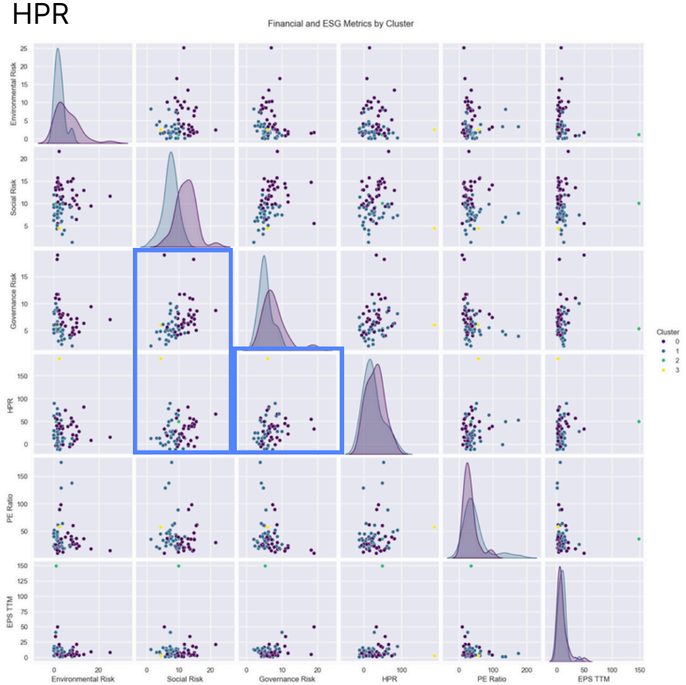
Results

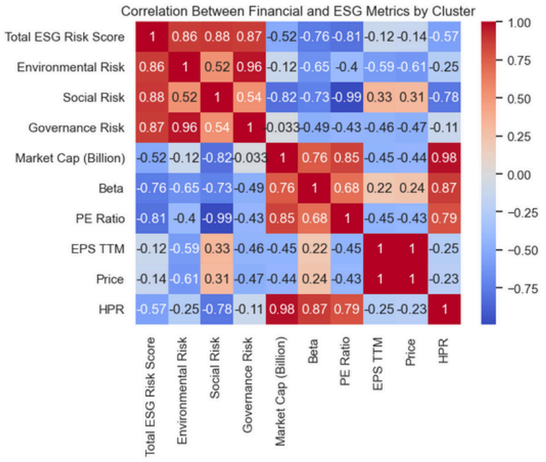
- High correlation observed between Social and Governance Risks
- Certain clusters show elevated Social Risk aligned with higher HPR
- One cluster features outliers with significantly higher EPS TTM
- No clear linear trend between ESG risks and PE Ratio or EPS TTM across all clusters.
- Correlation heatmap: strong Env and Soc Risk alignment (r ~ 0.96), moderate Social Risk correlations with EPS TTM (r ~ 0.33) and Price (r ~ 0.31), and minimal direct correlation between ESG metrics and HPR

Cluster	Total ESG Risk Score	Environmental Risk	Social Risk
0	26.108824	6.085294	12.202941
1	15.266667	2.515152	7.387879
2	16.400000	1.100000	10.000000
3	12.900000	2.500000	4.400000

Cluster	Governance Risk	Market Cap (Billion)	Beta	PE Ratio
0	7.826471	437.427941	0.919912	31.547350
1	5.363636	467.096667	1.049894	46.428247
2	5.300000	175.010000	1.385000	35.460970
3	6.000000	3552.520000	1.657000	57.110237

Cluster	EPS TTM	Price	HPR
0	9.238235	245.427353	30.210249
1	9.675152	371.416667	24.430082
2	149.120000	5287.939941	49.458130
3	2.540000	145.059998	187.016281





Discussion

- ESG metrics are instructive, yet do not have any independent predictive power on stock prices.

- Reliable prediction requires a combination of ESG scores with financial indicators and external factors.

Challenges and Opportunities

Challenges

1. Weak Predictive Power of ESG Alone
2. Limited sample size and short data collection period
3. Residual diagnostics revealed heteroscedasticity and non-linear patterns that standard transformations didn't fully resolve

Opportunities

1. Integrate macroeconomic indicators, sentiment data (news, social media), and sector-specific indices
2. Exploring non-linear models, robust regressions, or neural networks
3. Including more companies/expand into international

Thank You